



SQL CLASS NOTE – MODULE 11:

ORDER BY

1. Introduction

The **ORDER BY** clause is used to **sort query results** based on one or more columns.

- By default, sorting is **ascending (ASC)** (from small → large, A → Z).
- You can also sort in **descending (DESC)** (from large → small, Z → A).

👉 Example:

```
SELECT * FROM Students
ORDER BY Age ASC;    -- youngest first
```

```
SELECT * FROM Students
ORDER BY Age DESC;   -- oldest first
```

2. Syntax

```
SELECT column1, column2, ...
FROM table_name
ORDER BY column1 [ASC|DESC], column2 [ASC|DESC];
```

- ASC = Ascending (default).
- DESC = Descending.
- You can order by **multiple columns**.

3. Practical Examples with SchoolDB

Example 1: List students ordered by Age (youngest → oldest)

```
SELECT FirstName, LastName, Age
FROM Students
ORDER BY Age ASC;
```

Example 2: List students ordered by Age (oldest → youngest)

```
SELECT FirstName, LastName, Age
FROM Students
ORDER BY Age DESC;
```

Example 3: Order courses alphabetically by CourseName

```
SELECT CourseName, CourseCode
FROM Courses
ORDER BY CourseName ASC;
```

Example 4: Order students first by GradeLevel, then by Age (both ascending)

```
SELECT FirstName, LastName, GradeLevel, Age
FROM Students
ORDER BY GradeLevel ASC, Age ASC;
```

Example 5: Order enrollments by date (latest first)

```
SELECT StudentID, CourseID, EnrollmentDate
FROM Enrollments
ORDER BY EnrollmentDate DESC;
```

Example 6: Order students by LastName alphabetically

```
SELECT FirstName, LastName
FROM Students
ORDER BY LastName ASC;
```

Example 7: Order courses by Teacher name (descending)

```
SELECT CourseName, Teacher
FROM Courses
ORDER BY Teacher DESC;
```

4. Why Use ORDER BY ?

- Makes results **more readable**.
- Useful in reports (e.g., top students, latest enrollments).
- Helps to **rank or prioritize** data.

5. Real-Life Use Cases

- **Schools:** Get oldest students in a class.
- **E-commerce:** Sort products by price (low → high).
- **Banks:** Show transactions from latest to oldest.
- **Hospitals:** Sort patients alphabetically by name.

6. Classwork (Practice)

 Try these queries:

1. Display all students ordered by **FirstName** alphabetically.
2. Show students ordered by **GradeLevel (ASC)** and then by **Age (DESC)**.
3. List courses ordered by **CourseCode (ascending)**.
4. Show enrollments ordered by **EnrollmentDate (oldest first)**.
5. Display students ordered by **Age (DESC)**.
6. Order students by **LastName (A–Z)**.
7. Order students by **Gender first**, then by **FirstName**.

7. Challenge Question

 Write a query to display the `FullName` , `Age` , and `GradeLevel` of students, ordered by:

1. **GradeLevel (ASC)**
2. **Age (DESC)**
3. **FirstName (ASC)**

 That's the **expanded Module 11: ORDER BY**.

Perfect  Let's solve the **Module 11 Classwork + Challenge** step by step with SQL queries.

Solutions to Module 11: ORDER BY

1. Display all students ordered by **FirstName** alphabetically.

```
SELECT FirstName, LastName, GradeLevel
FROM Students
ORDER BY FirstName ASC;
```

2. Show students ordered by **GradeLevel (ASC)** and then by **Age (DESC)**.

```
SELECT FirstName, LastName, GradeLevel, Age
FROM Students
ORDER BY GradeLevel ASC, Age DESC;
```

3. List courses ordered by **CourseCode (ascending)**.

```
SELECT CourseName, CourseCode, Teacher
FROM Courses
ORDER BY CourseCode ASC;
```

4. Show enrollments ordered by **EnrollmentDate (oldest first)**.

```
SELECT StudentID, CourseID, EnrollmentDate
FROM Enrollments
ORDER BY EnrollmentDate ASC;
```

5. Display students ordered by **Age (DESC)**.

```
SELECT FirstName, LastName, Age
FROM Students
ORDER BY Age DESC;
```

6. Order students by **LastName (A-Z)**.

```
SELECT FirstName, LastName, GradeLevel
FROM Students
ORDER BY LastName ASC;
```

7. Order students by **Gender first**, then by **FirstName**.

```
SELECT FirstName, LastName, Gender
FROM Students
ORDER BY Gender ASC, FirstName ASC;
```



Challenge Question

👉 Display the `FullName`, `Age`, and `GradeLevel` of students, ordered by:

1. **GradeLevel (ASC)**
2. **Age (DESC)**
3. **FirstName (ASC)**

```
SELECT CONCAT(FirstName, ' ', LastName) AS FullName,
        Age,
        GradeLevel
FROM Students
ORDER BY GradeLevel ASC, Age DESC, FirstName ASC;
```

📌 This ensures:

- Students are grouped by `GradeLevel`,
- Within each grade, oldest students appear first,
- If two students are the same age, they are sorted alphabetically by `FirstName`.

✅ That completes **Module 11: ORDER BY**.

👉 Do you want me to now continue with **Module 12: Aggregate Functions (COUNT, SUM, AVG, MIN, MAX)** in the same expanded, practical style?